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Palaeomagnetism of the Börzsöny Mountains (Hungary)

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Summary—Mean directions of magnetization (29 normal and 31 reversed) were recorded for 60 magmatic localities of middle Miocene age from the Börzsöny Mountains (Hungary). The overall mean direction of RM, irrespective of polarity, is $D = 0{,}9°$; $I = 59{,}8°$; with $k = 8{,}3$ and $\alpha_{95} = 6{,}8°$. The coordinates of the corresponding geomagnetic north pole are $\Phi = 82{,}7°$, $\Lambda = 193{,}8$ with $\Delta p = 7{,}7°$ and $\Delta m = 10{,}2°$.

Key words: Palaeomagnetism of miocene rocks from Hungary; Miocene rocks from Hungary, Palaeomagnetism; Hungary, Palaeomagnetism of miocene rocks.

Introduction

This paper gives a concise description of the results of a palaeomagnetic study in the Börzsöny Mountains (Hungary). Details of the work have been reported earlier (MÁRTON and KIS, 1970–74).

The samples were collected from andesites and dacites of lower Helvetian to upper Tortonian ages (PANTÓ and MIKÓ, 1964). The distribution of the 87 localities sampled is shown on the map (Fig. 1).

On average, seven hand samples were taken from each locality. The measurements of remanence were carried out on 2 cm cubic specimens using a JR-2 spinner magnetometer.

A progressive AF-demagnetization curve was obtained for each sample. Generally, a peak field of 400 (250) Oe proved to be satisfactory to isolate the stable component of remanence if the latter existed.

When computing the site mean direction of magnetization we rejected all samples that showed unstable behaviour on AF-cleaning. Further, we rejected all sites but two for which Fisher's α_{95} (cone of confidence of the site mean direction of magnetization) was greater than 15° of arc. Finally, we rejected a locality if, irrespective of its accuracy, the site mean itself exhibited a continuous change on progressive demagnetization up to the peak field applied. On these grounds 27 localities had to be disregarded.

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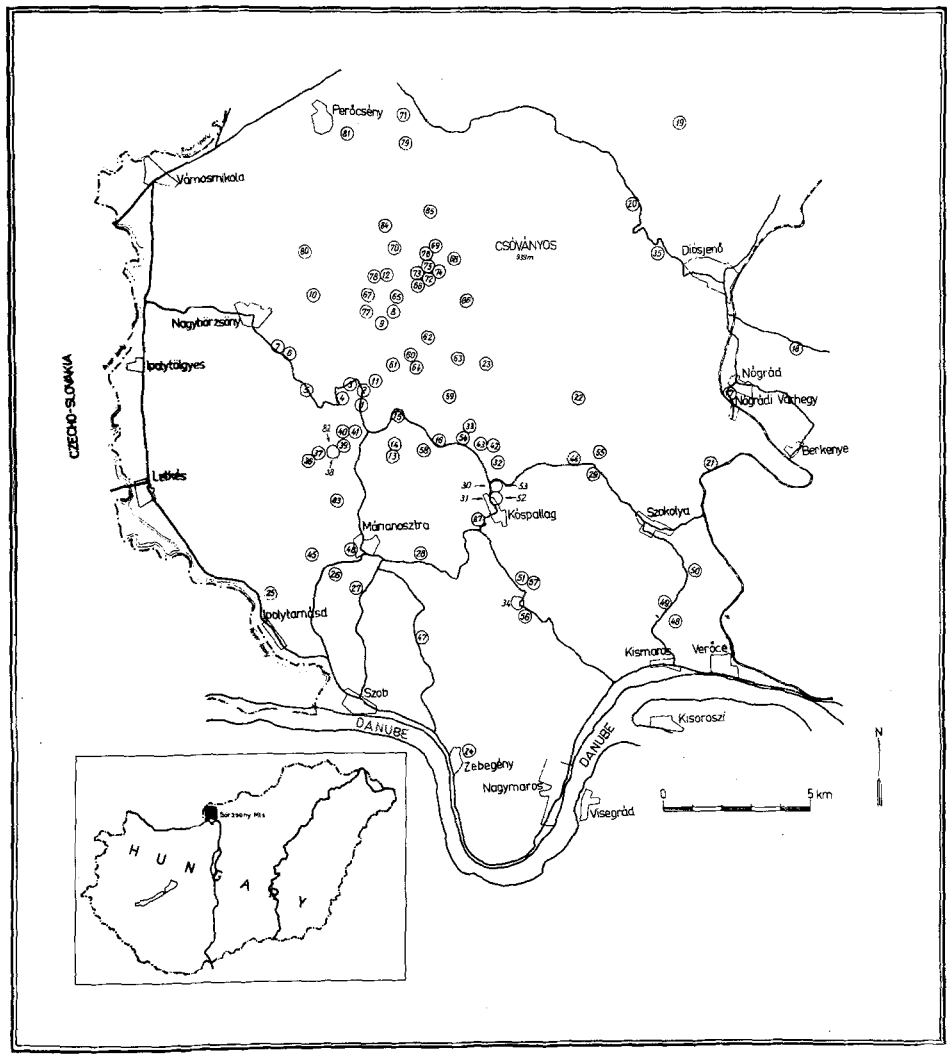
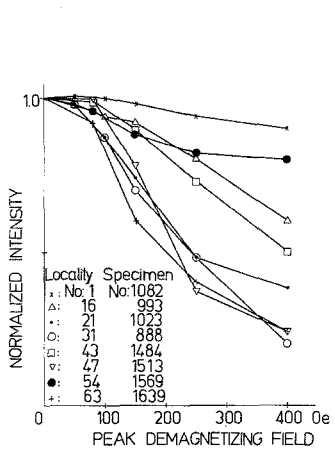


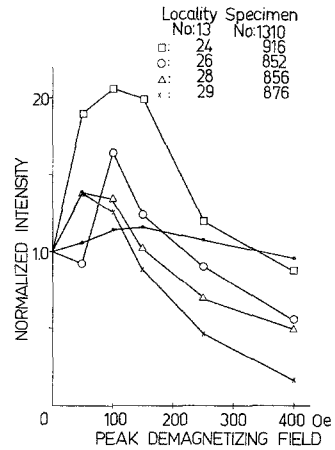
Figure 1
Map showing sampling localities in the Börzsöny Mountains.

Results and their analysis

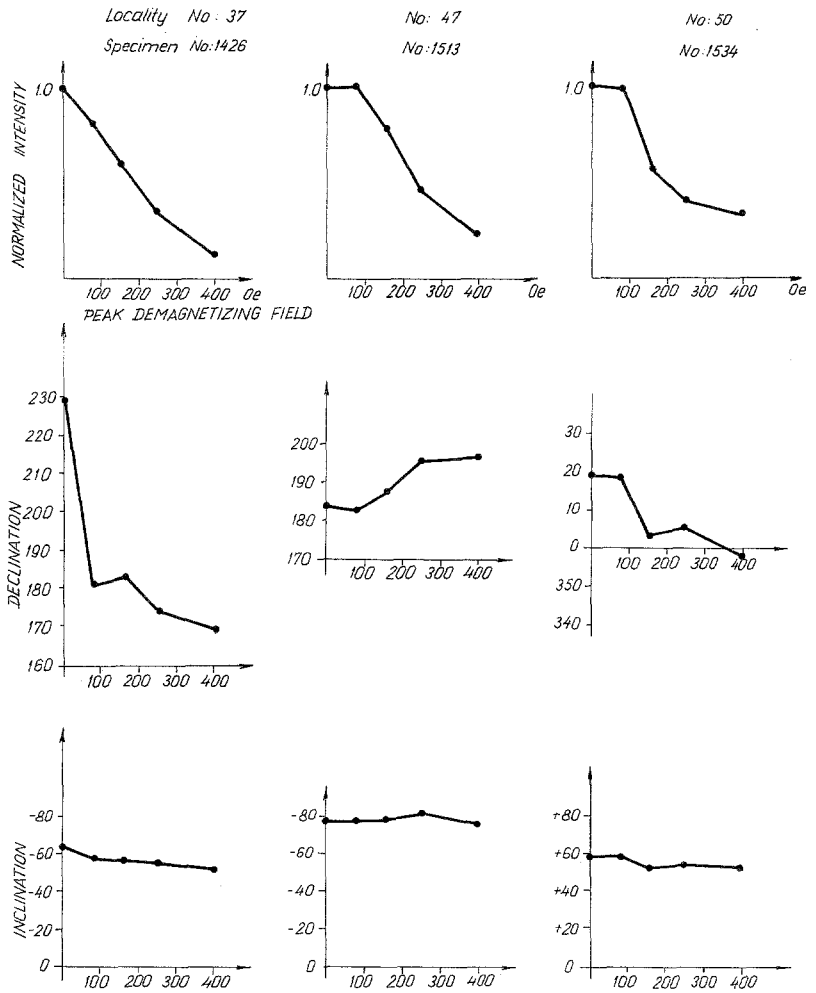
In Fig. 2 some representative AF demagnetization curves are presented. (a) is for samples containing no detectable secondary component. (b) is for samples of reversed magnetization containing, initially, a smaller or greater secondary magnetization of normal polarity. In (a) and (b) no appreciable change of the direction of NRM is detectable. (c) is for samples in which the removed secondary component is visible from the directional change. (d) is for samples, each one from a locality of con-


NORMALIZED INTENSITY
1.0
Locality Specimen
No: 1 No:1082
16 993
21 1023
31 888
43 1484
47 1513
54 1569
63 1639
0 100 200 300 400 Oe
PEAK DEMAGNETIZING FIELD

(a)


Locality Specimen
No:13 No:1310
24 916
26 852
28 856
29 876
NORMALIZED INTENSITY
20
1.0
0 100 200 300 400 Oe
PEAK DEMAGNETIZING FIELD

(b)


Locality No: 37
Specimen No:1426
No: 47
No:1513
No: 50
No:1534
NORMALIZED INTENSITY
1.0
100 200 300 400 Oe
PEAK DEMAGNETIZING FIELD
DECLINATION
230
220
210
200
190
180
170
160
30
20
10
0
350
340
INCLINATION
-80
-60
-40
-20
0
+80
+60
+40
+20

(c)

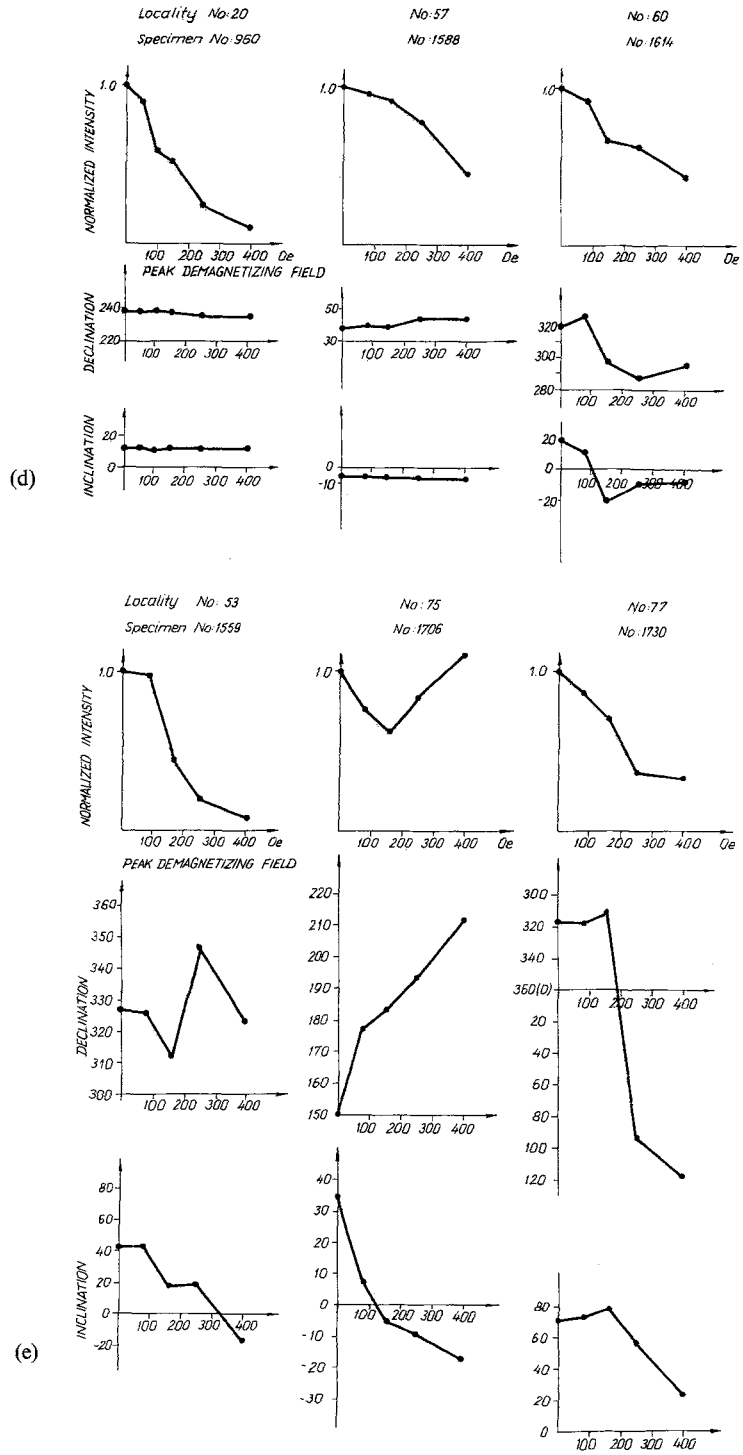


Figure 2
Some representative AF demagnetization curves for Börzsöny volcanics. For explanation see text.

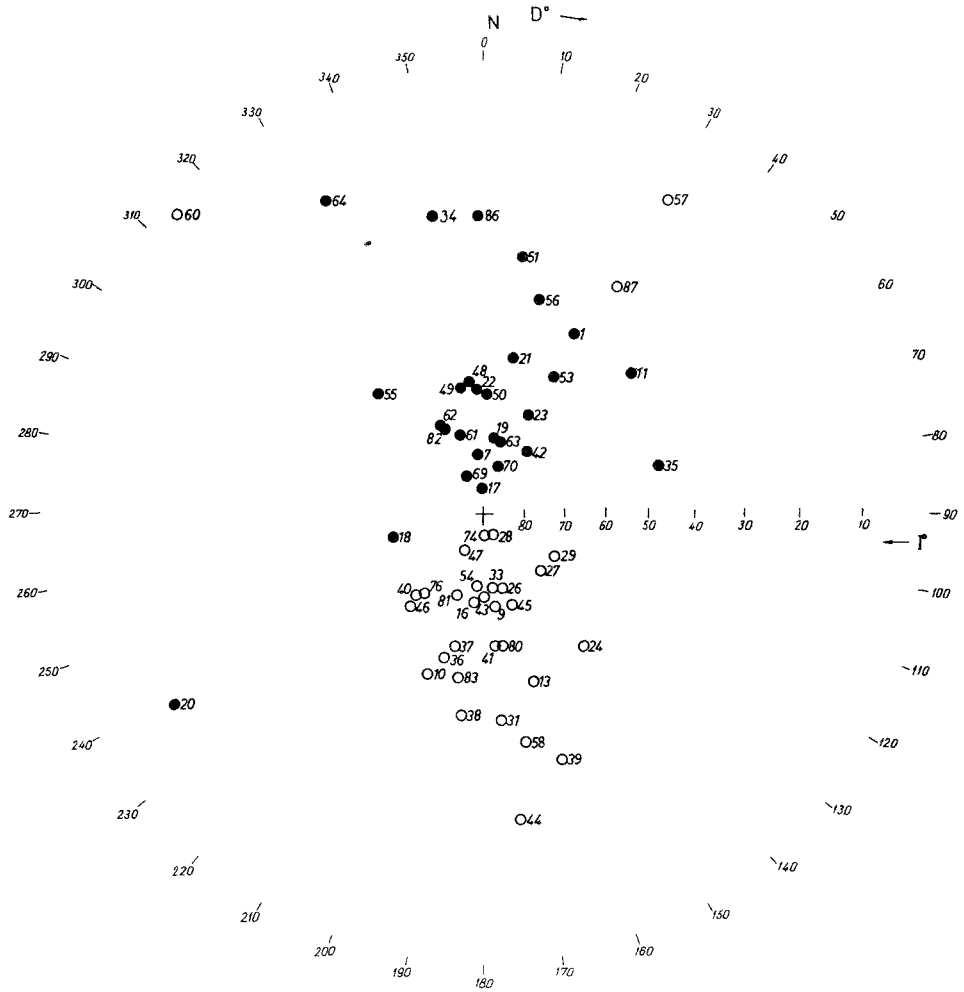


Figure 3
Stereographic projection of mean directions of magnetization in the Börzsöny Mountains. Full circles: positive (downward) inclination; hollow circles: negative (upward) inclination.

spiciously anomalous NRM direction. The curves clearly show that each sample contains a single, stable NRM. (e) is for samples in which no stable magnetization component is isolable up to 400 Oe peak field.

Stable magnetization was isolated in 60 localities. The distribution of the site mean directions of magnetization is shown in Fig. 3 and the relevant numerical values are listed in Table 1. The numerals in the fourth column headed by 'AF-demagnetization Oe' of Table 1 show the demagnetization stage in which the mean direction of magnetization was computed with the greatest precision (Fisher's k). Number 0 in

Table 1
Summary of primary palaeomagnetic data for the Börzsöny Mountains

Locality Number	Code/Name of Locality	N/N_0	AF Demagn. Oe	$D°$	$I°$	k	$\alpha_{95}°$
1	PM-1	6/6	0	27.5	+42.5	57.6	8.0
2	PM-2	6/6	150	27.9	+53.2	117.0	6.2
7	PM-6	4/13	400	354.9	+75.5	74.1	10.7
9	Rózsaszikla, outcrop	6/6	400	171.8	−67.2	15.3	17.6
10	Börzsöny-brook-quarry	9/9	250	199.2	−48.7	38.6	8.4
11	Kisirtáspuszta quarry	6/7	250	47.9	+42.8	275.0	4.0
13	Sóshegy, quarry	6/11	150	162.9	−47.8	25.0	12.3
16	Lóhegy, roadcut	6/6	250	184.9	−68.1	92.0	7.0
17	Nógrád, Castle Hill, quarry	6/6	250	359.0	+84.0	33.3	11.8
18	Somlyóhegy, quarry	10/10	150	255.2	+67.0	41.3	7.6
19	Kőember, quarry	12/12	250	8.6	+71.0	69.9	5.2
20	Málnapatak-quarry	14/14	100	237.9	+12.5	150.1	3.9
21	Rókahegy, quarry	8/8	150	11.1	+52.4	129.8	4.9
22	Bajdázóhegy, quarry	8/8	250	357.2	+60.0	141.0	4.7
23	Inóci-quarries	7/26	150	25.5	+63.6	234.0	3.9
24	Zebegény, Kerekhegy, quarry	6/6	150	142.5	−50.1	291.7	3.9
26	Csákhegy, Malomvölgy-quarry	7/7	250	164.8	−70.8	21.7	13.3
27	Csákhegy, White-quarry	8/8	100	134.2	−69.7	108.9	5.3
28	Czerina-quarry	9/9	250	153.2	−84.0	41.4	8.1
29	Paphegy, quarry	8/8	150	120.6	−69.2	35.4	10.5
31	Pereshegy, quarry	7/7	0	174.4	−40.9	127.7	5.7
32	Aranyoskút-quarry	3/3	0	335.5	+67.0	100.0	12.5
33	Pokolvölgy-quarry	6/6	150	170.3	−71.1	114.0	7.2
34	Kismaros-Kóspallag road, roadcut	8/8	100	350.1	+23.3	66.5	6.8
35	Závoz-quarry	6/8	150	75.3	+46.0	15.0	18.1
36	Szarvaskő, roadcut I	6/6	80	195.2	−54.5	51.3	9.4
37	Szarvaskő, roadcut II	4/4	160	192.1	−57.1	70.9	11.0
38	Szarvaskő, roadcut III	13/14	160	186.5	−42.5	63.7	5.2
39	Szarvaskő, roadcut IV	8/8	160	162.6	−31.2	31.9	10.0
40	Szarvaskő, roadcut V	6/11	400	219.0	−64.6	77.6	7.7
41	Szarvaskő, roadcut VI	10/10	160	174.1	−57.7	324.2	2.7
42	Ábrahámkert-quarry	9/9	250	36.4	+71.2	108.9	5.0
43	Pokolvölgy, outcrop	7/7	160	179.0	−68.8	63.5	7.6
44	Királyrét-Kóspallag road, quarry	6/6	160	172.7	−22.7	22.0	14.6
45	Brieszka-quarry	11/11	160	162.2	−66.7	29.7	8.5
46	Márianosztra, outcrop	5/5	250	217.9	−61.9	34.7	13.2
47	Bőszob, roadcut	8/8	400	206.3	−79.7	82.4	6.1
48	Börzsönyliget, quarry	10/10	250	354.3	+57.8	108.0	4.7
49	Börzsönyliget, outcrop I	5/5	250	349.5	+59.2	216.5	5.2
50	Börzsönyliget, outcrop II	10/10	160	1.9	+61.1	28.8	9.2
51	Pusztatorony I, outcrop	6/6	400	8.8	+31.6	23.4	14.2
54	Négyhányás, roadcut	10/10	250	184.7	−72.3	19.9	12.7
55	Királyrét, outcrop	7/7	160	318.5	+51.5	28.2	11.6
56	Korompa-brook, outcrop	4/8	160	15.3	+38.6	118.4	8.5
57	Pusztatorony II, outcrop	4/9	160	30.8	−13.0	45.7	13.7
58	Nagy Sashegy, roadcut	8/8	400	169.2	−36.2	67.9	7.4
60	Lajosforrás, outcrop	8/8	400	314.3	− 3.6	46.0	8.2

Table 1.—*contd.*

Locality Number	Code/Name of Locality	N/N_0	AF Demagn. Oe	$D°$	$I°$	k	α_{95}°
61	Szénáshegy, outcrop	7/7	250	344.1	+69.7	28.8	12.7
62	Korombérc, outcrop	6/6	80	334.3	+66.4	116.9	6.2
63	Nagyinóc, outcrop	6/8	400	13.3	+71.7	22.3	14.5
64	Ökrészkert, outcrop	6/6	400	333.8	+15.3	27.3	13.1
69	Csarna völgy, Mogyorósbérc outcrop	5/5	400	335.8	+80.3	39.7	12.3
70	Pintér-brook, Németút, outcrop	3/6	400	18.5	+78.0	96.7	12.6
74	Kuruc-brook, outcrop	7/7	160	175.3	−85.0	34.1	10.5
76	Ridge between Szoros brook and Kuruc-brook, outcrop	5/7	160/250	216.4	−65.9	67.4	8.2
80	Harmospuszta, Hegyeshegy, outcrop	6/6	400	171.6	−57.5	31.6	13.8
81	Perőcsény, Bedéktüzi-quarry	5/6	150	197.2	−69.1	251.0	4.8
83	Márianosztra, Bezina-quarry	6/6	150	188.5	−50.1	110.8	6.4
86	Nagyhideghegy, peak cliff	8/8	150	358.9	+23.8	115.3	5.2
87	Kóspallag, Tolvajkút outcrop	4/4	150	31.4	−29.9	102.2	9.1

Localities shown on the map (Fig. 1) but not referred to in Table 1 are not having stable magnetization.

Legend: N/N_0: number of samples used/collected; $D°$: site mean declination of magnetization; $I°$: site mean inclination of magnetization; k: precision (FISHER, 1953); α_{95}: semiangle of cone of 95% confidence (FISHER, 1953).

this column indicates that the site mean and its precision did not change appreciably through demagnetization.

The analysis of the directions of remanence during demagnetization suggests that a single component of remanence was isolated for each locality. Assuming a simple state of affairs, this remanence is most naturally identified with the primary magnetization.

In a more refined interpretation of the magnetization the possibility of the presence of a stable low temperature CRM should also be considered. We believe that setting the value of α_{95} relatively low for rejection is a safeguard that CRM affected localities are not used palaeomagnetically.

According to MERRILL (1976) the only instant that the presence of a low temperature chemical remanence can produce an erroneous magnetic direction is when it is formed predominantly during one magnetic field polarity. This case cannot be excluded entirely but it is more likely for a secondary CRM to be acquired during the whole life of the rock thereby to compose of oppositely-directed magnetizations and eventually to have a very moderate effect on the total remanence (MERRILL, 1975).

Twenty-nine normal and thirty-one reversed magnetizations were recorded. The site means in both groups exhibit a large dispersion. Fisher's precision measure (FISHER, 1953) is as low as $k = 8{,}3$ for the overall mean direction of RM.

Major dispersion sources may be as follows:

(i) Relative movements of localities subsequent to the aquisition of magnetization,

(ii) Secular variation,

(iii) Field reversals and excursions.

(i) There is no geologic observation suggesting an appreciable tilt for the Börzsöny area as a whole.

The lack of embedding sediments does not allow the measurement of the local tectonic position of the outcrops sampled. It is known, however, that the formations are horizontal in the eastern ranges, and it appears now that the dips are smaller than 10–20 degrees elsewhere (Z. BALLA, personal communication).

(ii) The average amplitude of secular variation during quiet periods of the geomagnetic field is 15–25° (TOICHIEV, 1976). However, secular variation seems to enhance in certain periods during the reversals and excursions of the geomagnetic field and its average amplitude attains 60° (TOICHIEV, 1976). Thus, it is very likely that tectonic movements and secular variation together are sufficient to account for the statistical scatter of the Börzsöny site mean directions.

(iii) As for the highly anomalous directions of site means of RM Nos 20, 57, 60, 87 we suggest that they reflect real field directions during field reversals. Otherwise each anomalous magnetization should be the resultant of at least two nearly antiparallel magnetizations of similar intensity, e.g., a TRM and a secondary CRM which, however, would probably have been discovered during AF-cleaning. (A single secondary CRM would not be of anomalous direction in late Tertiary times.)

Assuming that every source of dispersion causes random distribution of the site means of RM under discussion the following overall mean directions of magnetization and related virtual geomagnetic pole positions were computed (Table 2).

Note that the overall means of RM define, statistically, the same dipole axis for the normal and reversed polarity localities.

Table 2

Overall mean directions of magnetization for the Börzsöny Mountains

				$\varphi = 47.95°$		$\lambda = 18.96°$			
	N	$\overline{D}°$	$\overline{I}°$	k	$\alpha_{95}°$	$\Phi°$	$\Lambda°$	$\Delta p°$	$\Delta m°$
Normal polarity units	29	2.4	+57.1	7.4	10.5	79.6	188.3	11.2	15.4
Reversed polarity units	31	179.2	−62.2	9.1	9.1	−85.5	26.0	11.0	14.1
All units, irrespective of polarity	60	0.9	+59.8	8.3	6.8	82.7	193.8	7.7	10.2

Legend: φ, λ: mean geographic coordinates for the Börzsöny Mts.; $\overline{D}°$: overall mean declination of magnetization; $\overline{I}°$: overall mean inclination of magnetization; k: precision (FISHER, 1953); α_{95}: semiangle of cone of 95% confidence (FISHER, 1953); Φ, Λ: geographic coordinates of palaeomagnetic pole; Δp, Δm: semiaxes of oval confidence for palaeomagnetic pole position (FISHER, 1953).

An implication of the results

The oldest formations (intrusions and lavas) sampled along the east margin of the area (Fig. 1) are normally magnetized. Also a few localities representing the youngest

lavas are of normal magnetization. In between there must be at least one negative polarity zone resulting in a minimum of three polarity zones for the period of the Börzsöny volcanic activity.

At present a more refined palaeomagnetic subdivision of the interval in question is not feasible. However, attempts are being made to reconstruct the volcanic history of the area in detail (BALLA, 1976). As soon as the history is better known the existing polarity scale can be checked and extended.

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